Assessment rubric

How the final challenge is judged

Criterion	Beginning (1)	Developing (2)	Secure (3)	Excellent (4)
Follows the line	Leaves the line on most curves	Completes straight sections and gentle curves	Completes a full lap with two or more curves	Full lap including a sharp corner, smoothly
Build quality	Loose parts or wires	Works but wires near the wheels	Firm, tidy, sensors at the right height	Neat, labelled, easy to debug
Code	Copied, does not compile	Compiles with help	Compiles; pins and speeds named; comments	Tuned values explained in comments
Testing and fixing	No record of tests	One problem described	Problem, cause and fix described	Several changes tested one at a time, with results
Explaining	Cannot explain how it works	Explains sense–think–act	Explains the four rules	Explains why the robot behaves as it does on corners

Virtual pathway: “Follows the line” is judged from the recorded simulator runs (S-curve and sharp turns). Pass: 12 of 20 or more, with at least 2 in “Follows the line”.